Curriculum Vitae

Professor Aboagye Menyeh

Professor of Geophysics
Department of Physics, Kwame Nkrumah University of Science and Technology (KNUST)
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Contact Information

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Professional Summary

Professor Aboagye Menyeh is a distinguished geophysicist with over four decades of experience in research, teaching, and academic administration at KNUST. His pioneering work on the magnetic properties of monoclinic pyrrhotite (Fe_7S_8) has established him as an international authority in mineral magnetism, with 517+ citations and an h-index of 12. Beyond his research contributions, he served as the Foundation Provost of the College of Science (2004-2010), transforming the institution through strategic leadership, international partnerships, and infrastructure development. The College of Science building was named the "Aboagye Menyeh Complex" in 2015 in recognition of his transformative contributions.

Education

Ph.D. in Geophysics
University of Newcastle upon Tyne, United Kingdom (1987-1991)
Commonwealth Scholar
Dissertation: "The Magnetization Process of Monoclinic Pyrrhotite (Fe_7S_8)"
Supervisor: Prof. William O' Reilly

M.Sc. in Physics
Kwame Nkrumah University of Science and Technology, Ghana (1982-1984)
Specialization: Geophysics

B.Sc. (Hons) in Physics
Kwame Nkrumah University of Science and Technology, Ghana (1976-1980)
First Class Honours

Academic Positions

Position	Institution	Years
Professor of Geophysics	KNUST, Department of Physics	2008 - Present
Associate Professor	KNUST, Department of Physics	2000 - 2008
Senior Lecturer	KNUST, Department of Physics	1995 - 2000
Lecturer	KNUST, Department of Physics	1991 - 1995
Assistant Lecturer	KNUST, Department of Physics	1984 - 1987
Teaching Assistant	KNUST, Department of Physics	1980 - 1982

Administrative Leadership

Foundation Provost, College of Science (2004-2010)
Kwame Nkrumah University of Science and Technology

- Led the transformation from Faculty of Science to College of Science structure
- Secured over $800,000 in research funding and donations
- Established international partnerships with universities in Germany, UK, and across Africa
- Oversaw infrastructure development including new laboratories and research facilities
- Implemented academic programs that positioned the College as a leader in scientific research in West Africa

Dean, Faculty of Science (2001-2004)
Kwame Nkrumah University of Science and Technology

- Last Dean before transition to College system
- Developed strategic plan for structural transformation
- Strengthened research capacity and international collaborations
- Enhanced student training programs and graduate education

Research Interests

- **Mineral Magnetism:** Magnetic properties of monoclinic pyrrhotite (Fe_7S_8), domain structures, coercive force mechanisms
- **Geophysical Exploration:** Gold mineralization, groundwater prospecting, environmental geophysics
- **Rock Magnetism:** Paleomagnetism, magnetic minerals in geological materials
- **Applied Geophysics:** Electromagnetic methods, magnetic surveys, mineral prospectivity modeling

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Research Metrics

- **Total Citations:** 517+
- **h-index:** 12
- **i10-index:** 14

- **Years of Active Research:** 33 (1990-2023)
- **Publications:** 36+ peer-reviewed articles
- **Research Collaborations:** UK (Newcastle upon Tyne), Germany (Munich), Ghana, Nigeria, South Africa

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Selected Publications

Highly Cited Works

1. **Menyeh, A.**, & O' Reilly, W. (1991). The magnetisation process in monoclinic pyrrhotite (Fe_7S_8) particles containing few domains. *Geophysical Journal International*, 104(2), 387-399. [Citations: 120+]

2. **Menyeh, A.**, & O' Reilly, W. (1995). Magnetic hysteresis properties of fine particles of monoclinic pyrrhotite Fe_7S_8. *Journal of Geomagnetism and Geoelectricity*, 47(2), 173-184. [Citations: 95+]

3. **Menyeh, A.**, O' Reilly, W., & Soffel, H. C. (1997). Changes in magnetic properties of synthetic analogues of monoclinic pyrrhotite during heating. *Physics of the Earth and Planetary Interiors*, 103(3-4), 235-248. [Citations: 78+]

Recent Publications (2015-2023)

1. Wemegah, D. D., Fiandaca, G., Auken, E., Menyeh, A., & Danuor, S. K. (2023). Spectral time-domain induced polarisation and magnetic surveying—an efficient tool for characterisation of solid waste deposits in developing countries. *Near Surface Geophysics*, 21(1), 51-70.

2. Forson, E. D., Menyeh, A., Wemegah, D. D., Preko, K., & Manu, E. (2023). Mineral prospectivity mapping of gold mineralization in the Kibi-Winneba belt of Ghana based on Fuzzy analytical hierarchy process, Concentration-area (C-A) fractal model and prediction-area (P-A) plot. *Heliyon*, 9(5), e15641.

3. Deatanyah, P., Menyeh, A., Gyamfi, S., Adaramola, M. S., & Manu, E. (2021). Assessment of electromagnetic radiation from broadcast transmitters and mobile base stations in Ghana. *Scientific African*, 12, e00777.

Research Grants and Funding

Principal Investigator
Teaching and Learning Innovation Fund (TALIF) Grants (2000-2010)
Total Funding: $800,000

- Geophysical equipment acquisition
- Student training and capacity building
- Field research expeditions in Ghana
- Laboratory infrastructure development

Co-Investigator
Ghanaian-German Lake Bosumtwi Drilling Project (2004-2012)
International collaborative research on paleoclimate and impact crater geology

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International Collaborations

University of Newcastle upon Tyne, United Kingdom
Collaborative research on monoclinic pyrrhotite magnetization (1987-2000)
Multiple joint publications with Prof. William O' Reilly

Ludwig Maximilian University of Munich, Germany
Research on synthetic pyrrhotite analogues and thermal magnetic properties (1995-2002)
Joint publications with Prof. Heinrich C. Soffel

Lake Bosumtwi International Research Team
Ghanaian-German collaboration on drilling project (2004-2012)
Co-authored publications on geophysical surveys and paleomagnetic studies

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Graduate Student Supervision

Ph.D. Students Supervised

- Dr. David Dotse Wemegah (Completed 2021) - Now Lecturer, KNUST Physics Department
- Dr. Emmanuel Daanoba Forson (Completed 2023) - Geophysics Researcher
- Dr. Peter Deatanyah (Completed 2021) - Environmental Geophysics Researcher

M.Phil. Students Supervised

Over 20 M.Phil. students in geophysics, mineral exploration, and environmental physics (1995-2023)

Awards and Honors

Year	Award/Honor	Institution
2015	Aboagye Menyeh Complex Naming	KNUST College of Science
2008	Promotion to Full Professor	KNUST
2004-2010	Foundation Provost Appointment	KNUST College of Science
2001-2004	Dean of Faculty of Science	KNUST
2000	Promotion to Associate Professor	KNUST
1995	Promotion to Senior Lecturer	KNUST
1987-1991	Commonwealth Scholarship	UK Government

Professional Memberships

- Geophysical Society of Ghana
- African Geophysical Society

- International Association of Geomagnetism and Aeronomy (IAGA)
- Commonwealth Scholarship Alumni Network

Teaching Experience

Undergraduate Courses Taught

- Introduction to Geophysics
- Exploration Geophysics
- Rock Magnetism and Paleomagnetism
- Electromagnetic Theory
- Mathematical Methods in Physics

Graduate Courses Taught

- Advanced Geophysical Methods
- Mineral Magnetism
- Geophysical Data Processing and Interpretation
- Research Methods in Geophysics

Community and Institutional Service

- **Member, University Council** - KNUST (2010-2015)
- **Chair, College Academic Board** - College of Science (2004-2010)
- **Member, University Appointments and Promotions Committee** - KNUST (2008-2015)
- **Reviewer** - Multiple international geophysics journals
- **External Examiner** - Universities in Ghana, Nigeria, and South Africa

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Research Impact

Professor Menyeh’s work on monoclinic pyrrhotite has had lasting impact on the field of mineral magnetism. His experimental studies on domain structures, coercive force mechanisms, and magnetization processes established new theoretical frameworks that continue to be cited decades after publication. His applied research on gold exploration and groundwater prospecting has contributed to Ghana’s natural resource sector, with practical applications in mineral exploration and water resource management.

His administrative leadership transformed KNUST’s College of Science into a world-class institution, with enhanced research infrastructure, international partnerships, and academic programs that continue to benefit students and faculty. The naming of the Aboagye Menyeh Complex stands as a permanent testament to his institutional legacy.

Languages

- English (Fluent)
- Twi (Native)

References

Available upon request

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